

Detecting Health Misinformation on Social Networking Sites Using Large Language Models and Deep Learning-based Natural Language Processing

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Abstract— Health misinformation on social networking sites (SNS) is a critical issue, particularly during health crises like the COVID-19 pandemic. The spread of inaccurate health information can lead to severe outcomes, including reduced vaccine uptake and decreased trust in healthcare. This paper introduces a system using large language models (LLMs) and deep learning-based natural language processing (NLP) techniques to detect and mitigate health misinformation on SNS. The model incorporates transformer-based architectures, zero-shot (ZS) and few-shot (FS) learning, and prompt engineering to classify health-related content as true or false. A key feature is the integration of a knowledge graph that enhances verification capabilities. Applied to both English and Arabic datasets, the system's performance is evaluated using accuracy, F1-score, BLEU, and METEOR metrics, demonstrating its multilingual effectiveness in real-time misinformation detection.

Keywords— Health Misinformation, Large Language Models (LLMs), Deep Learning, Natural Language Processing (NLP), Zero-Shot Learning, Few-Shot Learning, Prompt Engineering, Social Networking Sites (SNS).

I. INTRODUCTION

Health misinformation on social networking sites (SNS) has become a critical issue, especially during public health crises like the COVID-19 pandemic. Misinformation can lead to severe consequences, including reduced vaccine uptake, increased public confusion, and a decline in trust in healthcare systems. The rapid spread of inaccurate health-related information underscores the need for efficient detection and mitigation strategies.

Recent advancements in Natural Language Processing (NLP) have introduced large language models (LLMs), such as GPT-4, which offer powerful tools for understanding and generating language across multiple domains. LLMs, equipped with transformer-based architectures, have demonstrated remarkable success in tasks like text classification, sentiment analysis, and misinformation detection, particularly through zero-shot (ZS) and few-shot (FS) learning. These techniques enable models to classify unseen content without requiring extensive labeled datasets, making them ideal for the dynamic and evolving nature of health misinformation.

This paper presents a system that leverages LLMs, specifically GPT-4, along with deep learning-based NLP techniques, to detect health misinformation on SNS. A unique aspect of this work is the integration of knowledge graphs to capture relationships between health-related entities, enhancing the system's verification capabilities. By applying this system to both English and Arabic datasets, we aim to create a multilingual solution that addresses misinformation detection in diverse linguistic contexts.

The system's effectiveness is evaluated through various metrics, including accuracy, F1-score, BLEU, and METEOR, demonstrating its potential to improve real-time health misinformation detection and support public health initiatives.

A. Scope of the Project

This project aims to address the pervasive issue of health misinformation on social networking sites (SNS), with a specific focus on developing a multilingual detection system that incorporates both English and Arabic. The system is designed to benefit a wide range of users, including institutions and individuals across Saudi Arabia and globally, who rely on SNS for exchanging information, browsing content, and staying informed on health-related topics.

The core objective of this project is to protect users from the potential dangers of false health information that can mislead and harm individuals. Misinformation on SNS can cause serious public health consequences, and thus, developing an effective detection mechanism is critical. By utilizing advanced technologies such as large language models (LLMs) and deep learning-based natural language processing (NLP), the proposed system will provide users with a reliable tool to identify and mitigate health misinformation.

The detection model integrates both zero-shot and few-shot learning techniques to ensure that it can accurately classify misinformation without extensive training on specific datasets. This multilingual capability, which supports both English and Modern Standard Arabic (MSA), will broaden the applicability of the system, ensuring it can serve diverse linguistic communities. Ultimately, this project aims to contribute to safer, more informed social media usage by providing accurate, real-time detection of health misinformation.

B. Expected Outcomes

This research is expected to yield several significant contributions to the field of health misinformation detection, particularly through the application of large language models (LLMs) and deep learning-based natural language processing (NLP). The following outcomes are anticipated:

1. **Advanced Detection Model:** Develop a robust AI-driven system capable of accurately identifying health misinformation on social networking sites (SNS) in real-time. This system will be designed to handle both English and Arabic content, addressing the gaps in current misinformation detection approaches.
2. **Improved Health Literacy:** By promoting the dissemination of accurate health information, the system will enhance public awareness and empower users to make informed health decisions, thereby improving overall health literacy.
3. **Multilingual Misinformation Detection:** The system will provide a novel approach to detecting misinformation in both English and Modern Standard Arabic (MSA), significantly extending the reach of misinformation detection tools to non-English-speaking communities.
4. **Policy Recommendations:** Insights gained from this research will offer valuable guidance for policymakers and health authorities in formulating strategies to combat health misinformation, contributing to safer online environments and more effective public health communication.

C. Research Questions

We aim to address the following research questions:

- 1) Can Large Language Models (LLM) and deep learning-based natural language processing and reasoning over knowledge graphs improve the detection of health misinformation?
- 2) What are the optimal ways to represent and reason over knowledge graphs to detect health misinformation?
- 3) How does the proposed approach compare to current state-of-the-art methods for detecting health misinformation?
- 4) How can Prompt Engineer improve misinformation detection in the health domain?

II. LITERATURE REVIEW

The rapid proliferation of health misinformation on social networking sites (SNS) has become an alarming issue, particularly during public health crises like the COVID-19 pandemic. Several studies have explored various machine learning (ML) and natural language processing (NLP) techniques to address this challenge, with a predominant focus on English-language content. Despite this, there is a significant gap in applying these methods to non-English languages, especially Arabic, which represents a major linguistic group within SNS users. This review discusses key approaches and identifies areas where further development is needed, particularly with the advent of large language models (LLMs) and the expanding role of deep learning techniques.

A. English-Language Misinformation Detection Approaches

A vast body of work has focused on detecting misinformation in English. Groza et al. (2020) introduced an ontology-based approach to counteract misinformation, particularly addressing health-related false claims during the COVID-19 pandemic. Their method leveraged medical ontologies to reason about and validate health claims. While effective in domain-specific misinformation, the approach was limited to English datasets and struggled with the real-time detection needed on dynamic SNS platforms[1].

Kou et al. (2022) proposed a CEA-COVID framework, employing Natural Language Explanations (NLE) to enhance explainability in the detection of COVID-19 misinformation on SNS. This approach significantly improved detection accuracy, utilizing crowdsourced expert knowledge to train the model. However, its dependence on domain-specific knowledge limited its generalizability and scalability, especially in multilingual contexts[2].

Building on these, Upadhyay et al. (2023) developed the Vec4Cred system, using a CNN-BiLSTM network to analyze web content reliability. Their approach demonstrated superior accuracy for detecting health misinformation in English, but it lacked adaptability to other languages, particularly Arabic, and did not exploit the benefits of LLMs[3].

B. Arabic Misinformation Detection: A Critical Gap

Arabic, spoken by over 400 million people, remains significantly underrepresented in misinformation detection research. Alkhalifa et al. (2021) identified the unique linguistic challenges specific to Arabic, including its rich morphology and complex syntax, which hinder the direct application of machine learning (ML) and natural language processing (NLP) models trained on English datasets. Notably, few studies address Modern Standard Arabic (MSA), the formal written form of Arabic widely used on social media, leaving a substantial gap in the development of effective misinformation detection models for the Arab world [25].

Alkhalifa et al. (2021) further emphasize the difficulty of applying existing ML models to Arabic content, particularly MSA, due to the language's unique features, such as intricate morphology and a vast lexicon. These challenges lead to lower model accuracy when applied to Arabic datasets, as most NLP models were not originally designed to handle these linguistic complexities. Moreover, the field of Arabic misinformation detection remains in its early stages, with limited models specifically adapted to the characteristics of Arabic misinformation [25].

Additionally, the scarcity of publicly available annotated datasets further complicates efforts to develop robust models for Arabic misinformation detection. Current methodologies often rely on the heuristic translation of English datasets into Arabic, introducing translation errors and compromising model performance. This review highlights the critical need for developing language-specific models tailored to MSA and regional Arabic dialects to ensure accurate and effective misinformation detection in the Arabic-speaking world.

C. The Role of Large Language Models in Misinformation Detection

The introduction of large language models (LLMs), such as BERT and GPT, has revolutionized natural language processing (NLP) tasks by significantly enhancing context

understanding and language generation capabilities. Brown et al. (2020) demonstrated that LLMs, particularly GPT-3, excel in few-shot and zero-shot learning, enabling them to perform complex tasks without the need for extensive training data. This capability makes LLMs particularly effective in misinformation detection, where the rapidly evolving nature of misinformation and the scarcity of labeled data present significant challenges [26].

D. Combining Knowledge Graphs with LLMs

A novel advancement in misinformation detection is the integration of knowledge graphs with large language models (LLMs). Knowledge graphs provide structured representations of health-related entities and relationships, enabling models to reason more effectively about the veracity of health claims. Cui et al. (2020) introduced the DETERRENT model, which utilized a knowledge-guided graph attention network to enhance the detection of healthcare misinformation. Their system significantly outperformed traditional methods by incorporating external knowledge, demonstrating the effectiveness of hybrid models in misinformation detection [4].

The integration of LLMs with knowledge graphs presents a promising avenue for multilingual misinformation detection. This approach could significantly improve the model's reasoning capabilities, particularly in critical domains such as healthcare, where misinformation can have serious consequences.

III. METHODOLOGY

This project utilizes advanced large language models (LLMs) for the detection of health misinformation on social networking sites (SNS), without relying on specific datasets or traditional preprocessing techniques. Instead, it focuses on leveraging the capabilities of GPT-4, LangChain, and Chain-of-Thought (CoT) reasoning through zero-shot and few-shot prompting methods. The system is designed to dynamically evaluate and compare claims with existing studies, providing accurate and detailed misinformation assessments.

A. No Dataset and Preprocessing:

Unlike conventional machine learning projects, this system does not involve the collection, annotation, or preprocessing of specific datasets. Instead, it operates using the pre-trained capabilities of GPT-4, which has been trained on extensive corpora covering diverse topics, including health-related information.

Because the large language model can comprehend language patterns, nuances, and context, there is no need for text cleaning, tokenization, or other typical NLP preprocessing steps. This eliminates the need for handling raw data and significantly reduces the time and resources usually spent on data preparation.

B. Configure Connectivity with OpenAI LLM:

Once the decision to bypass traditional dataset usage and preprocessing is established, the next step involves configuring connectivity with OpenAI's GPT-4 large language model. This process includes the following technical configurations:

1. **API Integration:** The system establishes a secure connection with OpenAI's API, enabling seamless interaction with GPT-4. The API key is securely

stored and managed to ensure that the system can continuously access the model for prompt generation and misinformation detection tasks.

2. **Rate Limiting and Access Control:** Proper rate-limiting mechanisms are put in place to avoid hitting API request limits, ensuring that the system can handle large volumes of misinformation claims without interruption. Access control is configured to restrict API usage to authorized users and systems.
3. **Logging and Monitoring:** Real-time logging and monitoring are implemented to track API calls and model responses, allowing the team to analyze performance, response accuracy, and any anomalies in the detection process.
4. **Error Handling:** A robust error-handling system is implemented to ensure smooth recovery from issues such as API timeouts or service unavailability, ensuring high availability and reliability for the misinformation detection system.

Once the connectivity with OpenAI's GPT-4 LLM is configured, the system is capable of interacting with the model to process user queries, evaluate misinformation, and generate responses.

C. Leveraging Large Language Models (LLMs):

The core of the methodology revolves around GPT-4, one of the most advanced large language models available. It is employed to perform misinformation detection in both English and Arabic, using sophisticated prompting techniques:

- 1) **Zero-Shot Prompting:** GPT-4 is capable of evaluating claims and misinformation without being trained on domain-specific datasets. In zero-shot prompting, the model makes predictions or classifications on new information that it hasn't seen before. This method is especially useful for addressing emerging health misinformation, which often spreads rapidly on social media.
- 2) **Few-Shot Prompting:** In some cases, a small number of labeled examples are provided to fine-tune the model's response. Few-shot prompting enables the system to adjust and generalize across health-related topics based on minimal input data.

D. LangChain Integration:

The LangChain framework is integrated to enhance the model's reasoning and decision-making by connecting it to external tools and real-time data sources. This allows the system to fact-check claims against reliable sources, such as health databases, scientific literature, and verified news reports, adding robustness to the model's assessments by grounding them in up-to-date information.

LangChain Agents play a pivotal role in orchestrating interactions with external APIs, data sources, and verification systems. These agents enable the model to perform complex, multi-step tasks such as gathering, analyzing, and synthesizing information to provide accurate conclusions about health misinformation.

Key features of LangChain agents include:

1. **ReAct (Reasoning and Acting):** Agents autonomously reason through tasks, plan actions, and execute them, such as querying a health database to verify a medical claim and delivering a conclusion.
2. **Conversational Tree-of-Thought:** Agents simulate dynamic conversations, following a step-by-step reasoning process to evaluate complex claims involving multiple layers of ambiguity.
3. **Autonomous Agent Simulations:** Agents simulate decision-making processes to assess various types of health misinformation, improving the model's adaptability to different contexts.
4. **Prompt-Level Visibility:** Agents respond to prompts in a context-aware manner, ensuring accurate and relevant information is provided based on the user's intent.

These agents significantly enhance the system's ability to fact-check and verify claims in real-time, ensuring thorough evaluation before presenting the final output. This functionality is essential for dynamically responding to misinformation across platforms.

E. Chain-of-Thought (CoT) Reasoning:

Chain-of-Thought reasoning is employed to further enhance the model's ability to provide detailed and logical explanations. This reasoning technique allows the model to break down complex health misinformation claims into smaller, sequential steps, ensuring a thorough evaluation process. By guiding the model to follow a structured reasoning pathway, CoT improves the model's ability to explain its decision-making process, which is crucial for providing users with transparent and understandable results.

This approach is particularly effective in health misinformation detection, where a claim may involve nuanced medical terminology or intricate scientific concepts. CoT reasoning helps ensure that each aspect of the claim is examined in depth, improving the accuracy and reliability of the final assessment.

F. Fine-Tuning LLMs for Domain-Specific Tasks:

In addition to zero-shot and few-shot prompting, fine-tuning is employed to further optimize the performance of the LLMs for health misinformation detection. Fine-tuning involves adjusting the weights and parameters of a pre-trained model to better align with domain-specific requirements.

The fine-tuning process in this project focuses on optimizing GPT-4 for handling health-related misinformation, ensuring that the model can process and accurately assess claims from a variety of medical and health-related domains. This is particularly useful in improving the model's sensitivity to subtle nuances in medical terminology, claims of treatment efficacy, or vaccine-related misinformation.

Fine-tuning is conducted by training the LLM on a carefully curated subset of health-related text that includes misinformation and fact-checked claims. This allows the model to better understand the unique challenges posed by health misinformation, such as the use of technical jargon, conflicting expert opinions, and rapidly evolving information during health crises like pandemics.

G. Prompt Structure for Misinformation Detection:

To effectively guide GPT-4 in the evaluation and detection of health misinformation, a structured prompt format is employed. This format ensures that the model elicits detailed and comprehensive responses by covering all critical aspects of a claim. The structure includes the following components:

- **Claim:** The health-related statement or information being analyzed, which may be presented by the user or extracted from social media.
- **Source:** The origin of the claim, identifying whether it stems from a website, a social media post, or an individual, and assessing the credibility of the source.
- **Fact Check:** Verification of the claim against authoritative and reliable data sources, such as official health organizations (e.g., WHO, CDC), ensuring the accuracy of the information presented.
- **Logic:** Evaluation of the reasoning behind the claim, determining whether the logical structure is sound or if there are any flaws in the argument.
- **Language:** A linguistic analysis of the claim, focusing on identifying any signs of misleading or biased wording that could contribute to the spread of misinformation.
- **Context:** An examination of the broader context in which the claim appears, considering relevant trends in misinformation and other external factors that may influence the interpretation of the claim.
- **Harm:** An assessment of the potential negative impact or harm that could result from the dissemination of the misinformation, particularly in terms of public health or safety.
- **Conclusion:** A final assessment of the claim's validity, categorizing it as true, partially true, or false, accompanied by a reasoned explanation that justifies the model's evaluation.

This structured prompt format ensures that the model conducts a thorough and multidimensional evaluation of each claim. It addresses not only the factual accuracy but also the broader implications, such as context and harm, ensuring a logically consistent and comprehensive analysis of health-related misinformation.

H. Evaluation and Comparison:

The performance of the system is evaluated by comparing its results to findings from other studies in the field of misinformation detection. Since no specific dataset is used for training, the system's output is measured against established benchmarks in health misinformation research, including accuracy in claim verification, depth of explanation, and the clarity of reasoning.

Additionally, by utilizing zero-shot and few-shot prompting, the system is tested for its ability to adapt to new types of misinformation without requiring extensive retraining. This adaptability is crucial for addressing evolving health misinformation trends on social media.

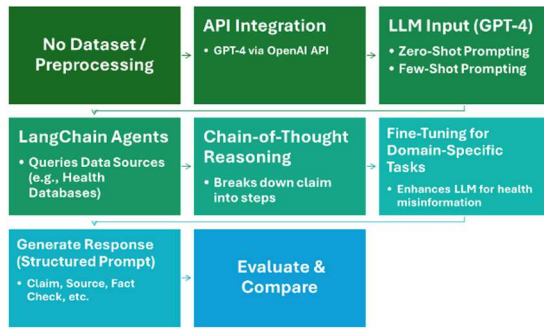


Figure 1. Proposed Methodology

IV. RESEARCH MATERIALS AND METHODS

This section outlines the materials, methods, and resources required for the development and evaluation of a system designed to detect health misinformation on social networking sites (SNS). The research leverages cutting-edge large language models (LLMs), advanced frameworks such as LangChain, and structured reasoning approaches to ensure accurate detection across multiple languages, specifically English and Arabic.

A. Research Materials:

a) Large Language Model (GPT-4):

The core of this research is based on OpenAI's GPT-4, a state-of-the-art large language model known for its capability to perform complex natural language processing (NLP) tasks. GPT-4 is selected for its ability to process health-related information through zero-shot and few-shot learning techniques.

- Platform: OpenAI GPT-4
- Capabilities: Zero-shot learning, few-shot learning, natural language understanding in English and Arabic, among other languages.

b) LangChain Framework

The LangChain framework is used to facilitate the model's interaction with external knowledge sources. By integrating LangChain, the system can access real-time data and use it to fact-check claims, ensuring up-to-date and accurate health misinformation detection.

- Platform: LangChain framework
- Features: Autonomous agents, real-time API querying, knowledge graph integration.

c) LangChain Agents:

LangChain agents serve as autonomous entities responsible for querying external databases and verifying the validity of claims by interacting with real-time sources like health databases.

- Key Features: Autonomous task execution, real-time querying, agent-driven decision-making.

d) API Integration for GPT-4 Connectivity

The system is integrated with OpenAI's API to ensure seamless communication with GPT-4. The API configuration includes secure access, rate limiting, and logging for real-time interaction with the model.

- Platform: OpenAI API
- Configurations: Secure access, rate limiting, logging, error handling.

B. Research Methods

The methodology is structured to test the system's ability to detect and evaluate health misinformation claims by leveraging LLM capabilities, LangChain integration, and structured reasoning techniques.

a) No Dataset and Preprocessing

This research does not rely on traditional datasets for model training. Instead, the system uses GPT-4's pre-trained capabilities to evaluate claims in real-time through zero-shot and few-shot prompting, eliminating the need for text cleaning or tokenization.

b) Zero-Shot and Few-Shot Prompting

Zero-shot and few-shot prompting techniques are employed to process and classify claims. These techniques allow the system to handle emerging misinformation trends without extensive training data or labeled datasets.

c) Chain-of-Thought (CoT) Reasoning

Chain-of-Thought reasoning breaks down complex claims into logical steps. This method enables the model to analyze claims based on logical consistency, context, and potential harm, providing detailed explanations for each assessment.

d) Prompt Structure for Misinformation Detection

The system evaluates misinformation claims using a structured prompt, which includes:

- Claim: The health-related statement being evaluated.
- Source: The origin of the claim.
- Fact Check: Verification against reliable sources.
- Logic: Assessment of the logical validity of the claim.
- Language: Review of language used in the claim for bias or misleading elements.
- Context: The broader context in which the claim is made.
- Harm: Assessment of potential harm caused by the claim.
- Conclusion: Final determination of the claim's validity.

e) LangChain Agent Interactions

LangChain agents facilitate real-time interactions with external sources for fact-checking, such as querying medical databases and cross-referencing claim details with reliable data.

f) Fine-Tuning

Fine-tuning GPT-4 is performed to improve its handling of health-related misinformation. Fine-tuning allows the model to better understand nuanced medical terms and health-related concepts, ensuring higher accuracy.

g) Evaluation Metrics

- Accuracy: The system's rate of correct classifications.
- F1 Score: Balance between precision and recall.
- BLEU and METEOR Scores: Evaluation of text generation quality, especially in multilingual contexts.
- Expert Review: Health experts validate the system's output for real-world accuracy.

C. Resources Required

The following resources are essential for the successful development, testing, and deployment of the health misinformation detection system:

a) Computational Resources

- **Cloud Infrastructure:** The system relies on cloud computing resources for running GPT-4 and managing LangChain agent operations. A scalable infrastructure like AWS or Google Cloud is required to handle real-time API calls and heavy processing loads.
- **GPU/TPU Instances:** High-performance instances, such as those with GPU (Graphics Processing Unit) or TPU (Tensor Processing Unit) support, are necessary for efficient model inference, particularly when handling large volumes of data and real-time requests.

b) OpenAI API Access

GPT-4 API Key: A subscription to OpenAI's API for accessing GPT-4. Depending on the usage, a sufficient quota of API calls is necessary to allow the system to handle high volumes of misinformation detection tasks.

c) LangChain Framework

- **LangChain Subscription/Setup:** LangChain integration requires access to its framework and APIs. Real-time querying and agent-based systems demand reliable connectivity and scalability.
- **External Knowledge Sources:** Integration with external health databases (such as WHO, CDC, or other medical sources) is necessary to ensure up-to-date fact-checking and verification.

d) Development Tools

- **Version Control:** Tools like Git or GitHub are required for code collaboration, version tracking, and system updates.
- **Integrated Development Environment (IDE):** Tools such as Visual Studio Code or PyCharm for code development and debugging.

e) Expert Collaboration

- **Health Experts for Validation:** Collaboration with medical professionals or public health experts is required to validate the system's outputs and ensure that the results align with real-world medical knowledge.

f) Testing and Validation Tools

- **Testing Frameworks:** Automated testing frameworks (e.g., PyTest, TensorFlow) to evaluate system performance and stability.
- **Benchmarking Tools:** Tools for evaluating the system's accuracy and comparing it with existing models for misinformation detection.

V. SYSTEM IMPLEMENTATION

The system implementation involved integrating several advanced components, including OpenAI's GPT-4, LangChain agents, and Chain-of-Thought (CoT) reasoning, to detect and assess health misinformation across multiple languages. The following sections outline the key steps involved in building and deploying the system.

A. API Integration with GPT-4

The system is built around GPT-4, accessed via the OpenAI API. The API integration enables real-time processing of claims using zero-shot and few-shot prompting techniques, without requiring domain-specific training data. Error-handling mechanisms, including rate-limiting and logging, ensure stable and continuous operation.

B. LangChain Agent Integration

LangChain agents are responsible for querying external health databases, such as WHO and CDC, to fact-check claims in real-time. The agents enhance the system's verification process by autonomously retrieving up-to-date data, ensuring that claims are cross-checked with authoritative sources.

C. Chain-of-Thought Reasoning

CoT reasoning was embedded to enable the system to logically break down complex health claims into a series of steps for thorough evaluation. This method ensures that the system's conclusions are well-reasoned and transparent.

D. Fine-Tuning GPT-4

To optimize GPT-4 for the detection of health misinformation, fine-tuning was performed using health-related datasets. This enhanced the model's ability to interpret medical terminology and assess the validity of claims, further improving its accuracy in this domain.

E. System Workflow

- **Claim Input:** Users submit health-related claims for verification.
- **Initial Processing:** GPT-4 processes the claim using structured prompting.
- **LangChain Agents:** Agents query external data sources for real-time verification.
- **CoT Reasoning:** Claims are evaluated through logical decomposition, ensuring a robust assessment.
- **Output Generation:** The system provides a detailed explanation of the claim's validity, including fact-checking results and potential harm assessment.

VI. TESTING & EVALUATION

This section presents the analysis and evaluation of the system's performance using OpenAI's GPT-4 model and a local large language model (LLM). We assess the models using translation benchmarks, namely METEOR and BLEU scores, to evaluate the responsiveness of zero-shot and few-shot learning techniques.

A. Fine-Tuning OpenAI

Figure 2 illustrates the loss chart for the fine-tuned OpenAI model. The significant reduction in error, approaching zero, indicates a high level of prediction accuracy. The convergence of training and validation curves confirms that the model is well-calibrated, with minimal overfitting or underfitting. To further improve performance, data preprocessing techniques (such as normalization and handling missing values) could be applied.



Figure 2. OpenAI loss chart

B. Accuracy for OpenAI Fine-Tuning

Figure 3 compares the accuracy and F1 scores for few-shot and zero-shot learning techniques. The results demonstrate an enhancement in the model's classification performance due to the advanced learning strategies applied, improving decision-making accuracy.

```
{'zs': {'accuracy': 0.87, 'f1': 0.77}, 'fs': {'accuracy': 0.86, 'f1': 0.8}}
```

Figure 3. Accuracy and F1 Score for OpenAI Fine-Tuning

C. Evaluating OpenAI Responsiveness

a) Zero-Shot and Few-Shot Response Evaluation

We evaluated the zero-shot and few-shot responses of the model using the same set of health-related statements. Figures 4, 5 and 6 provide the health statements and references used for both zero-shot and few-shot evaluations in English and Arabic, Figures 7 and 8 illustrate the prompts employed in the evaluations.

```
# Define health info prompts
health_info_prompts = [
    "Statement: 'Drinking hot water with lemon every morning can cure diabetes.'",
    "Statement: 'A ketogenic diet can reverse type 2 diabetes.'",
    "Statement: 'Vitamin C supplements can prevent the common cold.'",
    "Statement: 'Coconut oil can help you lose weight.'",
    "Statement: 'Yoga can cure chronic back pain.'",
    "Statement: 'Green tea extracts can improve brain function.'",
    "Statement: 'Intermittent fasting can extend your lifespan.'",
    "Statement: 'Eating soy can cause cancer.'",
    "Statement: 'Magnetic bracelets relieve arthritis pain.'",
    "Statement: 'Alkaline water can neutralize acid in your bloodstream.'",
    "Statement: 'Eating eggs increases the risk of heart disease.'",
    "Statement: 'Gluten-free diets are healthier for everyone.'",
    "Statement: 'Artificial sweeteners are safer than sugar.'",
    "Statement: 'Detox diets remove toxins from the body.'",
    "Statement: 'Wearing glasses weakens your eyesight over time.'",
    "Statement: 'Drinking celery juice daily detoxifies your body.'",
]
```

Figure 4. Health Statements Used for Zero-Shot and Few-Shot Evaluation

```
# Add reference translations for BLEU score calculation
reference_translations_en = [
    "Drinking hot water with lemon every morning can cure diabetes.",
    "A ketogenic diet can reverse type 2 diabetes.",
    "Vitamin C supplements can prevent the common cold.",
    "Coconut oil can help you lose weight.",
    "Yoga can cure chronic back pain.",
    "Green tea extracts can improve brain function.",
    "Intermittent fasting can extend your lifespan.",
    "Eating soy can cause cancer.",
    "Magnetic bracelets relieve arthritis pain.",
    "Alkaline water can neutralize acid in your bloodstream.",
    "Eating eggs increases the risk of heart disease.",
    "Gluten-free diets are healthier for everyone.",
    "Artificial sweeteners are safer than sugar.",
    "Detox diets remove toxins from the body.",
    "Wearing glasses weakens your eyesight over time.",
    "Drinking celery juice daily detoxifies your body.",
]
```

Figure 5. English References

```
reference_translations_ar = [
    "شرب المياه الدافئة مع الليمون كل صباح يمكن أن يعالج مرض السكري.",
    "يمكن لنظام الكيتو أن يعكس مرض السكري من النوع 2.",
    "يمكن لمكملات فيتامين سي أن تمنع نزلات البرد.",
    "يمكن لزيوت جوز الهند أن تساعد على فقدان الوزن.",
    "اليوغا يمكن أن تعالج آلام الظهر المزمنة.",
    "مستخلصات الشاي الأخضر يمكن أن تحسن وظائف الدم.",
    "يمكن أن يبطئ الميالح المتقطع حياتك.",
    "تناول الصويا يمكن أن يسبب السرطان.",
    "الأشاور المغناطيسية تخفف من آلام التهاب المفاصل.",
    "يمكن للمياه القلوية أن تحيد الحمض في مجرى الدم.",
    "تناول البيض يزيد من خطر الإصابة بأمراض القلب.",
    "النظام الغذائي الخالية من الجلوتين أكثر صحة للجميع.",
    "المكملات الصناعية أكثر أماناً من السكر.",
    "الحمية الغذائية لإزالة السموم تزيل السموم من الجسم.",
    "ارتداء النظارات يضعف بصرك بمرور الوقت.",
    "شرب عصير الكرفس يومياً يزيل السموم من جسمك.",
]
```

Figure 6. Arabic References

```
# Zero-shot task prompt
zero_shot_prompt = """Task: Evaluate the validity of health claims based on the information provided. Analyze each statement according to specified criteria and conclude if the claim is real or fake, with reasoning.

Criteria for Evaluation:
\n- Claim: Identify the central claim of the statement.
\n- Source: Assess the credibility of the source.
\n- Fact Check: Verify with reputable sources.
\n- Logic: Determine logical consistency.
\n- Language: Examine for neutrality or bias.
\n- Context: Consider the broader context.
\n- Harm: Evaluate potential harm if the claim is false.
\n- Conclusion: Decide if true, false, or further investigation needed.

Please analyze the following statements:
"""
```

Figure 7. Zero-Shot prompts

```
# Few-shot task examples
few_shot_examples = [
    {
        "statement": "Drinking hot water with lemon every morning can cure diabetes.",
        "claim": "Drinking hot water with lemon cures diabetes.",
        "fact_check": "No medical sources support this; not recognized by WHO or CDC.",
        "logic": "Lacks scientific basis; diabetes management is complex.",
        "language": "Sensational, promising a 'cure.'",
        "context": "Beliefs in natural remedies, missing crucial diabetes management context.",
        "harm": "Could mislead into neglecting proper medical treatments.",
        "conclusion": "Likely false, potentially harmful."
    },
    {
        "statement": "Regular exercise can help prevent heart disease.",
        "claim": "Regular exercise prevents heart disease.",
        "fact_check": "Supported by numerous studies and health organizations.",
        "logic": "Scientifically proven benefits for heart health.",
        "language": "Neutral and informative.",
        "context": "Overall lifestyle recommendations for heart health.",
        "harm": "Promotes healthy behavior, no harm.",
        "conclusion": "Likely true, well-supported by evidence."
    }
]
```

Figure 8. Few-Shot prompts

The performance of zero-shot and few-shot prompts is summarized in Tables I and II, showing BLEU and METEOR scores for both English and Arabic.

TABLE I. ZERO-SHOT BLEU AND METEOR SCORES

Prompt	English BLEU	English METEOR	Arabic BLEU	Arabic METEOR
Prompt 1	0.4883	0.1445	0.5225	0.1090
Prompt 2	1.6596	0.1803	0.9937	0.1332
Prompt 3	1.8532	0.2060	2.1786	0.1867
Prompt 4	3.2595	0.2474	3.6719	0.2989
Prompt 5	2.1887	0.2182	1.8037	0.1951
Prompt 6	2.4671	0.2406	0.3996	0.0819

TABLE II. FEW-SHOT BLEU AND METEOR SCORES

Prompt	English BLEU	English METEOR	Arabic BLEU	Arabic METEOR
Prompt 1	6.8350	0.3566	0.9928	0.1968
Prompt 2	5.2264	0.3953	1.5127	0.2481
Prompt 3	0.2437	0.0609	0.4335	0.0570
Prompt 4	2.9333	0.2090	1.4599	0.1776
Prompt 5	2.1887	0.2182	1.8037	0.1951
Prompt 6	5.2264	0.3953	1.5127	0.2481

From these results, it is evident that few-shot learning significantly improves the BLEU and METEOR scores, particularly in the English responses. However, the performance varied in Arabic prompts, and further improvement will focus on expanding the translation train set.

b) Drug Interaction Testing

Figure 9 illustrates the user interaction process with the drug interaction service, where a user inputs drug-related information and submits it for analysis via OpenAI. The analyzed results are then returned to the user, as depicted in Figure 10.

MEDICINES WITH GPT

Home / Medical articles / About / Contact / Hello user111@gmail.com / LOGOUT

Home / Send to GPT

Send to AI

What are the drug interactions between diabetes and high blood pressure?

SEND

ANS FROM AI

1. Diuretics (water pills) used to treat high blood pressure can increase blood sugar levels, which can be a concern for people with diabetes. 2. Beta blockers, commonly used to treat high blood pressure, can mask symptoms of low blood sugar in people with diabetes. 3. Angiotensin-converting enzyme (ACE) inhibitors and angiotensin receptor blockers (ARBs) have been shown to improve kidney function in people with diabetes, but they can also increase potassium levels in the blood. This can be a concern for people taking potassium-sparing diuretics or certain other medications that can also increase potassium levels. 4. Nonsteroidal anti-inflammatory drugs (NSAIDs) used to treat pain and inflammation can decrease the effectiveness of certain diabetes medications, potentially leading to high blood sugar levels. 5. Corticosteroids, used to treat a variety of conditions, can also raise blood sugar levels and may interfere with the action of insulin in people with diabetes. 6. Certain antibiotics, such as fluoroquinolones, have been shown to increase blood sugar levels in people with diabetes. 7. Aspirin, commonly used to prevent heart disease, can also increase blood sugar levels in people with diabetes. It is important for people with both diabetes and high blood pressure to carefully monitor their blood sugar levels and discuss any possible drug interactions with their healthcare provider.

Figure 9. Drug Interaction Service Process

Figure 10. User Response for Drug Interaction Service

VII. CONCLUSION

This paper presents a system that leverages GPT-4, a state-of-the-art large language model, to detect health misinformation on social networking sites. By incorporating zero-shot and few-shot learning along with knowledge graphs, the system effectively verifies health-related claims in both English and Arabic, offering a multilingual solution. The evaluation of the system using metrics such as accuracy, F1-score, BLEU, and METEOR demonstrates its capability to detect misinformation in real-time and its potential to enhance public health efforts.

This work contributes to the ongoing fight against health misinformation by offering a scalable and adaptable solution that addresses linguistic and contextual challenges. Future research will focus on further fine-tuning the model for domain-specific health misinformation and exploring additional language support.

ACKNOWLEDGMENT

The authors would like to express their sincere gratitude to King Faisal University and the College of Computer Science and Information Technology for their continuous support throughout this research. Special thanks to our supervisor, Dr. Hafiz Farooq Ahmed, from the Department of Computer Science, College of Computer Sciences and Information Technology, King Faisal University, Al-Ahsa, Saudi Arabia, for his invaluable guidance and expertise, which were instrumental in shaping the direction of this project.

We also acknowledge the contribution of Dr. Jamil Hussain, from the Department of Computer Science, Sejong University, Seoul, Korea, whose insightful feedback and assistance were vital to the development of this work.

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